

SELVA KARUPPASAMY M

AI Backend Engineer

✉ selvakaruppasamym@gmail.com ☎ 6379638115 📍 Bengaluru, Karnataka 🌐 selva-karuppasamym
🔗 SelvaKaruppasamyM

ABOUT ME

Backend Engineer with over 4 years of experience specializing in Python Tooling and high-performance System Architecture at Zentron Labs. Proven track record of bridging product vision with technical execution through cross-functional collaboration, delivering scalable, enterprise-grade solutions. Skilled in modern AI-driven development methodologies (TDD/SDD) and focused on driving measurable business value through optimized backend infrastructure.

WORK EXPERIENCE

Associate Software Developer 07/2022 – Present
Zentron Labs Pvt Ltd

- Collaborated with Product Owners to define technical requirements and delivered interactive Figma prototypes, accelerating stakeholder alignment by validating workflows prior to development and reducing initial feature deployment time by an estimated 1 week.
- Architected a scalable Python backend for a Computer Vision-as-a-Service (CVaaS) platform utilizing FastAPI and SQLAlchemy, capable of supporting 50 concurrent asynchronous tasks and handling 100+ MB of image data.
- Modernized legacy compute functions by implementing optimized Python modules via CFFI, reducing image processing latency by 50% and increasing overall system throughput to 15 transactions/frames per second."

SKILLS

Languages & Frameworks Python (FastAPI, Flask, SQLAlchemy), SQLite, MySQL, PostgreSQL, NoSQL, CFFI.	AI & Automation OpenCode Agent, Playwright MCP, TDD, SDD.
System & Design System Design, Microservices, Figma, REST API, Git, SVN, CI/CD Pipelines, WSL.	Vision & Device control LabVIEW, NI-VISA, NI-IMAQdx.

PROJECTS

CVaaS Infrastructure Development
Built a high-performance Python backend featuring asynchronous processing, PostgreSQL database integration, and real-time health monitoring for computer vision tasks.

Industrial Configurator Architecture
Designed a modular framework for an industrial configurator integrating C++ DLLs, reducing new feature integration time from 1 week to 3 days while minimizing technical debt.

AI-Powered Full-Stack MVP
Developed a real-time web application from scratch using Elixir, Phoenix LiveView, and TDD/SDD methodologies with OpenCode Agent. Integrated Playwright MCP to facilitate advanced AI-driven development and testing workflows.

EDUCATION

Bachelor of Electrical and Electronics Engineering <i>Alagappa Chettiar Government College of Engineering and Technology</i> 🎓 CGPA: 8.6/10	07/2018 – 04/2022 TamilNadu, India
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